

PBTRM — Prediction Basis Traceability Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Basis Standard (PBAS)

Operational Layer: Prediction Basis Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Basis Traceability

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction Basis Traceability

Canonical Definition

Prediction Basis Traceability Module (PBTRM) defines the governance conditions under which the complete informational foundation of a prediction remains identifiable, linked, historically reconstructable, version-aware, and distinguishable across changes in inputs, sources, relevance states, sufficiency states, and subsequent information availability.

PBTRM establishes the historical continuity of Prediction Basis.



← preserves the relationship between:

the prediction that was made

and:

the informational basis that existed when that prediction was made.

PBTRM prevents later information, updated sources, revised datasets, changed relevance determinations, or improved informational sufficiency from silently rewriting the basis of an earlier prediction.

Its foundational rule is:

Later Knowledge $\neq$ Original Prediction Basis

Operational Role

PBTRM serves as the fifth and final internal governance space of the Prediction Basis Standard (PBAS) .

It receives governed outputs from:

PIIM — Prediction Input Identification Module

PSMM — Predictive Source Mapping Module

SRVM — Signal Relevance Module

ISUM — Input Sufficiency Module

and binds them into a traceable Prediction Basis state.

The module asks:

Can the informational foundation used for this prediction be reconstructed later without contaminating it with information, sources, interpretations, or conditions that appeared only afterward?

The governed progression is:

Identified Inputs + Predictive Source Map + Relevant Signal Set + Sufficiency State $\rightarrow$ Basis Association $\rightarrow$ Traceability Structure $\rightarrow$ Historical State Preservation $\rightarrow$ Governed Prediction Basis Record

The resulting governed Prediction Basis becomes available to the next Parent Standard:

PAS — Prediction Assumption Standard.

Internal Module Flow

Prediction Input Identification $\rightarrow$ Predictive Source Mapping $\rightarrow$ Signal Relevance $\rightarrow$ Input Sufficiency $\rightarrow \rightarrow$ Prediction Basis Traceability

Current Position: Prediction Basis Traceability

Module Operational Space

PBTRM governs:

Prediction Basis identity; Prediction Basis record formation; prediction-to-basis association; input-to-basis association; source-to-input relationship preservation; relevance-state preservation; sufficiency-state preservation; basis-state timestamps; basis versions; basis snapshots; historical Prediction Basis; current Prediction Basis; basis change; basis revision; source-map change; input-state change; relevance change; sufficiency change; basis lineage; basis continuity; basis supersession; historical preservation; post-prediction information separation; retrospective contamination prevention; Prediction Basis Decay history; basis reconstruction; unresolved traceability gaps; and lifecycle handoff continuity.

Module Function

PBTRM prevents one of the most serious distortions in predictive governance:

judging an earlier prediction using an informational foundation that did not exist when the prediction was made.

Without traceability, organizations may unintentionally replace:

what the predictor knew then

with:

what the organization knows now.

A dataset may be updated.

A source may change.

A signal may later prove more important.

New evidence may emerge.

A methodology may receive additional information.

A Prediction Basis may become stronger or weaker.

None of these later developments should silently alter the historical informational state attached to the original prediction.

PBTRM therefore creates:

Prediction Basis State → Historical Preservation → Reconstructable Prediction Basis

Prediction Basis Identity

Each governed Prediction Basis should possess sufficient identity to distinguish it from other informational states.

Basis identity may be associated with:

a prediction; prediction version; prediction formation time; methodology version; predictive frame; informational state; basis version; other necessary lifecycle identifiers.

Prediction Identity ≠ Prediction Basis Identity

One prediction may possess multiple Prediction Basis states over time.

Prediction-to-Basis Association

A governed prediction must be associated with the Prediction Basis applicable at the point at which that prediction state was formed.

This relationship is fundamental.

Without it, later users may know the prediction output without knowing what information supported it.

Prediction Without Basis Association ≠ Fully Traceable Prediction

Prediction Basis Record

A Prediction Basis Record is the governed representation of the informational foundation associated with a prediction state.

It may reference:

identified inputs; input states; input timing; source relationships; source dependencies; relevant signals; relevance states; known unknowns; Prediction Basis gaps; sufficiency state; material limitations; basis timestamp; basis version; change history.

The record need not duplicate every underlying artifact.

It must preserve enough structured linkage to reconstruct the governed basis.

Basis Snapshot

A Prediction Basis Snapshot represents the governed informational state applicable at a defined point in the Prediction Lifecycle.

A snapshot may be formed at:

initial prediction formation; material basis change; prediction revision; prediction supersession; withdrawal; performance assessment; other material lifecycle transitions.

A snapshot preserves historical state.

It does not freeze operational systems permanently.

Historical Prediction Basis

The Historical Prediction Basis represents the informational foundation that actually applied to a prediction at a past point in time.

Historical basis must remain distinguishable from the current informational environment.

Current Information $\neq$ Historical Prediction Basis

Current Prediction Basis

Current Prediction Basis represents the informational foundation applicable to the current prediction state.

It may differ materially from the historical basis.

Both states may be legitimate.

They must not be conflated.

Current Prediction Basis $\neq$ Historical Prediction Basis

Prediction Basis Version

Where material changes occur, Prediction Basis may require version distinction.

A new version may be triggered by changes such as:

addition of material inputs; removal of material inputs; changed data state; changed source relationships; changed source access; changed relevance determinations; changed sufficiency state; material new information; basis restriction; basis expansion.

Same Prediction $\neq$ Same Prediction Basis Version by Default

Basis Change

Prediction Basis Change occurs where the informational foundation materially changes after an earlier governed basis state.

Changes may include:

new information; lost information; corrected information; source replacement; source dependency discovery; new relevant signal; relevance decay; resolved known unknown; new Prediction Basis Gap; sufficiency deterioration; sufficiency improvement.

PBTRM preserves the transition.

Basis Change Record

A material change should allow reconstruction of:

Previous Basis State $\rightarrow$ Change Event $\rightarrow$ New Basis State

The prior basis must not be overwritten.

Basis Update $\neq$ Historical Replacement

Prediction Basis Lineage

Prediction Basis Lineage represents the relationship among successive basis states associated with the same prediction lifecycle.

A lineage may appear as:

PB-1 $\rightarrow$ PB-2 $\rightarrow$ PB-3

where each state reflects a materially distinct informational foundation.

PBTRM preserves the sequence.

Input-State Traceability

An input may change over time.

PBTRM preserves which input state applied to which Prediction Basis version.

Examples include:

dataset version; sensor state; analyst assessment version; market value; model output; external report version.

Same Input Identifier $\neq$ Same Historical Input State

Source-State Traceability

Source relationships may also change.

A source that was independent at one point may later become dependent.

An organization may change ownership.

A predictor may later gain privileged information access.

PBTRM preserves which source structure applied to the historical Prediction Basis.

Relevance-State Traceability

A signal may be relevant at one point and irrelevant later.

PBTRM preserves:

original relevance state; conditions supporting that state; later reassessment; changed relevance state.

This prevents outcome knowledge from retrospectively changing what was considered relevant.

Sufficiency-State Traceability

Prediction Basis may move through states such as:

Sufficient

Conditionally Sufficient

Restricted

Partially Sufficient

Insufficient

PBTRM preserves which state applied to each governed basis version.

Current Sufficiency $\neq$ Historical Sufficiency

Temporal Traceability

Prediction Basis must be associated with relevant time.

Temporal traceability may include:

when information became available; when it was used; when basis state was formed; when it changed; when reassessment occurred.

This protects the prediction from retrospective information contamination.

Post-Prediction Information Separation

Information that becomes available after prediction formation must remain distinguishable from original Prediction Basis.

Such information may become relevant to:

monitoring; revision; supersession; outcome assessment; performance evaluation.

It must not silently become part of the historical basis.

Post-Prediction Information $\neq$ Original Prediction Basis

Retrospective Basis Contamination

Retrospective Basis Contamination occurs where later information is incorrectly represented as though it had supported an earlier prediction.

This may happen intentionally or unintentionally.

Examples include:

updated datasets replacing historical datasets; later news cited as original support; revised expert opinion replacing earlier assessment; hindsight reinterpretation of ambiguous signals; later source knowledge added retrospectively.

PBTRM requires these states to remain separated.

Retrospective Basis Drift

Retrospective Basis Drift occurs when the historical description of Prediction Basis gradually changes as later information, interpretation, or documentation is added.

The original informational state may become difficult to distinguish from later reconstruction.

PBTRM prevents this by preserving basis state and lineage.

Prediction Basis Decay Traceability

Prediction Basis may deteriorate over time.

PBTRM preserves:

when deterioration became identifiable; which informational elements changed; which relevance relationships weakened; which sufficiency conditions failed; resulting basis state.

Prediction Basis Decay $\neq$ Historical Basis Invalidity by Default

A basis may have been sufficient when formed and later become insufficient.

Correction and Historical Preservation

Information may later be discovered to have been incorrect.

PBTRM permits correction while preserving the historical fact that the original Prediction Basis contained the earlier information.

For example:

Historical Record: Input X = 52

Correction: Actual Input X should have been 25

Both facts may be governance-relevant.

Correction must not erase history.

Correction $\neq$ Historical Erasure

Source Correction

A source may later be found to have been incorrectly identified.

PBTRM permits source-map correction while preserving the original recorded state and subsequent correction.

This is necessary for auditability and historical reconstruction.

Prediction Basis Supersession

A new Prediction Basis version may supersede an earlier one for current predictive use.

The earlier basis remains part of prediction history.

Supersession $\neq$ Deletion

Basis Retirement

A Prediction Basis state may cease active use.

Retirement does not eliminate its historical relationship to the prediction states it supported.

Retired Basis $\neq$ Erased Basis

Basis Reconstruction

PBTRM should enable a later reviewer to answer:

What information was used? What information was unavailable? What information was excluded? Where did the information come from? Which sources were dependent? Which signals were considered relevant? What was the sufficiency state? What limitations were known? What was known at the time? What became known only later? Which basis version supported the prediction?

The purpose is not to recreate the future.

It is to reconstruct the informational conditions under which the prediction existed.

Traceability Granularity

Traceability should operate at a level sufficient to reconstruct material Prediction Basis relationships.

Not every raw data element must necessarily be individually archived where appropriate governed references, versions, datasets, or structured records can preserve the required state.

Traceability $\neq$ Unlimited Data Duplication

Reference-Based Traceability

PBTRM may preserve references to governed artifacts rather than copying them into a single Prediction Basis Record.

Examples include:

dataset identifiers; model-output identifiers; document references; source records; evidence references; simulation references; market-state references; system-state records.

The reference must be sufficiently persistent to support reconstruction.

Traceability Gap

A Prediction Basis Traceability Gap exists where a material historical relationship cannot be sufficiently reconstructed.

Examples include:

unknown input version; missing dataset snapshot; unknown source; lost source relationship; missing relevance determination;

unavailable sufficiency state; undocumented basis change; historical overwrite.

Traceability gaps should remain explicit.

Reconstruction Confidence

Historical reconstruction may sometimes be incomplete.

PBTRM permits confidence or completeness states where exact reconstruction cannot be guaranteed.

Potential states may include:

Complete

Substantially Complete

Partially Reconstructable

Materially Incomplete

Unable to Reconstruct

A reconstruction should not be represented as exact where material elements remain uncertain.

Historical Integrity

Historical integrity requires that Prediction Basis history preserve what was actually known, used, excluded, unavailable, and assessed at the relevant time.

It does not require that earlier information have been correct.

Historical Integrity $\neq$ Historical Correctness

Outcome Knowledge Separation

Once an outcome becomes known, that knowledge may create powerful hindsight effects.

PBTRM therefore requires Outcome Knowledge to remain distinguishable from original basis state.

Outcome Knowledge $\neq$ Original Prediction Basis

This distinction is essential for legitimate predictive performance assessment.

Performance Assessment Relationship

POFPS later compares predictions with governed outcomes.

PBTRM supplies the historical Prediction Basis needed to interpret that performance correctly.

Without it, performance review may confuse:

information advantage; hindsight; revised basis; changed source proximity; later knowledge with original predictive capability.

PBTRM does not perform predictive performance assessment itself.

Prediction Record Relationship

PRCS later governs the complete historical Prediction Record and Continuity.

PBTRM governs specifically the traceability of Prediction Basis.

Therefore:

Prediction Basis Record $\neq$ Complete Prediction Record

PBTRM contributes one governed historical layer to PRCS.

Governance Requirements

PBTRM requires that:

1. each material Prediction Basis be associable with the relevant prediction state; 2. basis identity remain distinguishable across material changes; 3. historical basis states remain reconstructable; 4. inputs remain traceable to the basis version in which they participated; 5. material source relationships remain historically preserved; 6. relevance states remain historically identifiable; 7. sufficiency states remain historically identifiable; 8. material basis changes create distinguishable historical states; 9. later information remain separate from original Prediction Basis; 10. updated information not silently overwrite historical inputs; 11. source changes not silently overwrite historical source structure; 12. relevance reassessment not rewrite historical relevance state; 13. later sufficiency improvement not inflate historical sufficiency; 14. corrections preserve prior recorded state where material; 15. Prediction Basis Decay remain traceable; 16. supersession not result in deletion; 17. retired basis states remain historically associated; 18. material traceability gaps remain explicit; 19. reconstruction uncertainty not be represented as certainty; 20. historical Prediction Basis remain usable for later performance and record governance.

Governance Logic

PBTRM applies the following governance logic:

Identified Prediction Inputs

Governed Predictive Source Map

Governed Relevant Signal Set

Governed Prediction Basis Sufficiency State

↓

Prediction Basis Identity Established

↓

Prediction-to-Basis Association Established

↓

Input / Source / Relevance / Sufficiency States Bound

↓

Basis Timestamp and Version Established

↓

Historical Basis Snapshot Preserved

↓

Material Basis Changes Detected

↓

New Basis State Created where required

↓

Previous Basis State Preserved

↓

Prediction Basis Lineage Maintained

↓

Post-Prediction Information Segregated

↓

Governed Prediction Basis Record

↓

PAS — Prediction Assumption Standard

Where historical relationships cannot be sufficiently reconstructed:

Prediction Basis Traceability Gap → Reconstruction / Clarification / Qualification / Historical Limitation Record

Minimum Implementation Framework

A minimum implementation of PBTRM must establish:

A. Prediction Reference

The prediction state associated with the Prediction Basis.

B. Prediction Basis Identifier

A sufficiently unique identifier for the governed Prediction Basis state.

C. Basis Version

The applicable Prediction Basis version.

D. Basis Timestamp

The point or interval at which the basis state applies.

E. Prediction Input References

The material Prediction Inputs associated with the basis.

F. Source Map Reference

The applicable governed source structure.

G. Signal Relevance Reference

The applicable governed relevance state.

H. Sufficiency State

The applicable PBAS sufficiency determination.

I. Known Gap Record

Material missing, unavailable, unresolved, or uncertain information.

J. Change Reference

Any material event creating a new Prediction Basis state.

K. Historical State Relationship

At minimum:

Initial / Current / Revised / Superseded / Retired

L. Predecessor / Successor Relationship

Where applicable, linkage between sequential basis states.

M. Post-Prediction Information Separation

Identification of information that became available only after the governed basis state.

N. Reconstruction State

At minimum:

Complete / Substantially Complete / Partially Reconstructable / Materially Incomplete / Unable to Reconstruct

O. Historical Preservation

The original governed basis state must remain preserved after later change.

Governance Outputs

PBTRM may produce:

Governed Prediction Basis Record; Prediction Basis Identifier; Prediction Basis Version; Prediction Basis Snapshot;

Prediction-to-Basis Association; Input-to-Basis Map; historical source-map reference; historical relevance-state reference; historical sufficiency-state record; Prediction Basis Lineage; basis-change record; basis-supersession record; basis-retirement record; Prediction Basis Decay history; post-prediction information record; retrospective-contamination flag; reconstruction-status record; Prediction Basis Traceability Gap Register; historical Prediction Basis package.

The primary governed output is:

Governed Prediction Basis Record

This completes PBAS and enables transition to:

PAS — Prediction Assumption Standard.

Critical Distinctions

Later Knowledge ≠ Original Prediction Basis

Prediction Identity ≠ Prediction Basis Identity

Prediction Without Basis Association ≠ Fully Traceable Prediction

Current Information ≠ Historical Prediction Basis

Current Prediction Basis ≠ Historical Prediction Basis

Same Prediction ≠ Same Prediction Basis Version by Default

Basis Update ≠ Historical Replacement

Same Input Identifier ≠ Same Historical Input State

Current Sufficiency ≠ Historical Sufficiency

Post-Prediction Information ≠ Original Prediction Basis

Prediction Basis Decay ≠ Historical Basis Invalidity by Default

Correction ≠ Historical Erasure

Supersession ≠ Deletion

Retired Basis ≠ Erased Basis

Traceability ≠ Unlimited Data Duplication

Historical Integrity ≠ Historical Correctness

Outcome Knowledge ≠ Original Prediction Basis

Prediction Basis Record ≠ Complete Prediction Record

Enterprise Application

PBTRM enables organizations to preserve the informational history of predictive systems across change.

This becomes critical where predictions depend on continuously changing:

datasets + APIs + AI outputs + model versions + market information + operational states + expert assessments + external intelligence.

Without Prediction Basis Traceability, organizations may later be able to reproduce a prediction output but still be unable to reconstruct:

what the system actually knew when it produced that output.

PBTRM therefore creates a durable historical connection between:

Prediction

and:

Informational State at Prediction Time

This supports legitimate later:

prediction revision; audit; review; performance analysis; accountability analysis; methodological comparison; historical reconstruction.

Use Case 1 — The Dataset Was Updated and the Original Prediction Can No Longer Be Reconstructed

A financial institution generates a credit-default prediction using a continuously updated external dataset.

Several months later, the prediction becomes subject to internal review.

The current dataset contains corrected values, updated company information, and historical revisions.

The prediction can still be reproduced using the current system, but the resulting output differs from the original.

The organization discovers that it never preserved which dataset state supported the original prediction.

The critical governance question is:

Can the institution legitimately explain or evaluate the original prediction when the informational foundation used at prediction time has been overwritten by later dataset states?

PBTRM requires the historical Prediction Basis state to remain reconstructable through governed version, timestamp, and input relationships.

The corrected dataset may govern current prediction.

It cannot silently replace historical Prediction Basis.

Current Information $\neq$ Historical Prediction Basis

Use Case 2 — A Prediction Looks Obvious After the Outcome Because Later Information Was

Added to Its History

A strategic forecasting team predicts that a major supplier will remain operational for the following six months.

Two months later, previously undisclosed financial problems become public.

The supplier subsequently collapses.

During post-event review, analysts reconstruct the case using all information currently available and conclude that the failure should have been obvious.

However, the most important financial information was not available to the forecasting team when the prediction was made.

The critical governance question is:

Can predictive performance be legitimately assessed using information that became available only after the original prediction had already been formed?

PBTRM preserves the original Prediction Basis separately from later information.

The later information becomes part of monitoring, revision, outcome, or performance history.

It does not become part of the original basis.

Outcome Knowledge $\neq$ Original Prediction Basis

Without this distinction, hindsight can be mistaken for predictive governance.

Canonical Boundary

PBTRM governs the identification, association, versioning, historical preservation, change lineage, reconstruction, and traceability of Prediction Basis across the Prediction Lifecycle. It preserves the historical relationships among Prediction Inputs, sources, relevance states, sufficiency states, and the prediction states they supported. It does not govern general evidence provenance, audit evidence, Prediction Assumptions, Prediction Formation, probability representation, prediction revision decisions, Operational Reality, predictive performance determination, or the complete Prediction Record governed later by PRCS.

PBTRM establishes:

what the prediction knew, from what informational state, and at what point in its lifecycle.

It does not determine:

whether what the prediction knew was true or whether the prediction was correct.

Canonical Closing Statement

A prediction cannot be fairly reconstructed, revised, or evaluated if its original informational foundation disappears behind what became known later.

PBTRM preserves the Prediction Basis as history — so PGA™ can always distinguish what was known then from what is known now.

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